

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer with experience designing, developing, and deploying production-grade computer vision systems for industrial and robotics applications for clients such as British Petroleum and Austin Industries. Strong background across the full model lifecycle, including problem framing, data curation, CVAT-based annotation, training, validation, optimisation, deployment, and iterative improvement. Hands-on experience with Python, C++, PyTorch, OpenCV, OpenVINO, and edge platforms. Proven track record building robust, maintainable inference pipelines using Git, CI/CD, DVC, and MLflow.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the full computer vision lifecycle for perception systems, from data curation to deployment and field maintenance using PyTorch and OpenVINO and improved F1 by 15% with deterministic 24/7 operation.
- Built end-to-end data ingestion, cleansing, training, and validation pipelines for multimodal sensor datasets using MLflow, DVC, Optuna and open-source CVAT cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Applied classical image-processing techniques to detect mud and sand splashes on camera lenses, reducing false positives by 13% and improving system reliability in harsh industrial conditions.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, improving fault isolation, increasing MTBF by 35%, and strengthening service continuity through logging, health checks, retries, and safe restarts.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers and strengthening maintenance and reliability during operation.
- Implemented Git and Jenkins-based continuous integration and validation workflows for edge inference modules for 24/7 deployments to improve release consistency across hardware targets.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy end-to-end production AI models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices using ONNX and OpenVINO.
- Re-architected the perception pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy-latency-memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50% with Keras and Intel OpenVINO.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulation-only continuation of the dissertation autonomy stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM plus LiDAR+RGB fusion SLAM and navigation.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM in 2D and 3D LiDAR across structured scenarios and implemented autonomous navigation with Nav2.
- Developed ROS2 nodes, launch files and parameterised configurations in C++ and Python and implemented tf2 frame trees with message synchronisation and debugged tf connectivity and transform timing issues.
- Implemented point cloud processing using PCL and Open3D for filtering, segmentation and registration to support mapping and localisation stability.
- Implemented an EKF-based localisation layer using robot_localization to fuse wheel odometry and IMU for smooth odom → base_link tracking when running LiDAR-only SLAM, improving motion continuity for Nav2.
- Containerised the stack with Docker on Linux and managed versioning in Git to ensure reproducible builds and consistent execution across development environments.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map, integrating multimodal LiDAR and radiological data for localisation and mapping in inspection scenarios leveraging the Clearpath Husky development platform.
- Evaluated RTAB-Map (LiDAR-only), LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation prior to real world deployment.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

SKILLS

- **Cloud, Edge & MLOps:** Apache Kafka, API design, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, Vision Transformer (ViT), YOLO
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Robotics Stack:** Gazebo (Harmonic), Nav2, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), ROS2 image_transport, tf2